

Modeling Animated Jumbo Floral Display on Disney's "Encanto"

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Figure 1: Final rendered images of selected areas of the jumbo floral display

ABSTRACT

Walt Disney Animation Studios' "Encanto" called for many dazzling musical numbers with special effects. Isabela's song "What Else Can I Do?" was one of the most technically challenging. In this sequence, Isabela finds freedom from the pressure and constraints of society's expectations, and lets her true self shine. Throughout this musical number, her environment represents her emotional transformation through the colorful, expressive, and hypnotizing flower patterns that appear on the walls of her bedroom. This effect required hundreds of thousands of flowers to change colors and move in kaleidoscopic patterns. This talk discusses the technical challenges encountered in bringing Isabela's room to life and the subsequent artistic and technical solutions.

CCS CONCEPTS

• Computing methodologies → Modeling methodologies.

KEYWORDS

procedural techniques, instancing, animated textures, effects

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1 INTRODUCTION

"Encanto" is a musical centering around an extraordinary family living in a magical house in Colombia. It called for many dazzling

musical numbers with special effects that reflect the special talents of the characters. Isabela, who was gifted with the magic of making plants and flowers grow at will, has a musical number that was envisioned to be performed in front of a jumbo 360-degree wall of flowers. Providing the required visuals to showcase the transformation of Isabela's character was integral to supporting this important moment in the story.

To realize this vision, artists and engineers had to work together creatively to find a solution that was both aesthetically appealing and practically feasible. Due to the scale and the complexity involved in the sequence, we opted to use Disney's XGen, our in-house procedural instancing tool as the key technical component in our workflow. However, the circumstances of this sequence provided unique challenges to the tools. The effect of the flowers on the walls coming to life required the procedural instances to not only move, but also change scales, orientations, colors, offsets, and various other properties. In the following section, we describe our method and processes to bring this jumbo floral display to life.

2 APPROACH

Utilizing our existing tools, the general idea was to use Disney's XGen to procedurally generate flower instances to cover the walls,



Figure 2: VisDev reference of the jumbo floral display

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with patterns, positions, scales, and motions driven by animated texture maps that were also procedurally generated using different techniques.

2.1 Animated Textures

Effects artists worked closely with the production designer in designing the unique animations and shapes for each shot in the sequence. The positioning and motion of each design was carefully crafted to enhance the energy and composition of each shot, as well as to underline different motifs and metaphors in the lyrics of the song. Once these animations and designs were finalized, animated textures were generated and handed off to the Look Development to control the generation and motion of the flower instances. These textures were split into three different passes in the RGB channels to allow each channel to be used for driving different properties of the procedural effects.

For the animation effects, animated images on the walls were indicated via shot-by-shot style frames for the sequence, as shown in Figure 3. The unwrapped wall geometry of Isabela’s room was huge compared with the very small area within the camera view needing an animated image. Because the camera was constantly moving, the unwrapped canvas area that needed animated images would appear and function like a small spotlight travelling across the walls. Only within that small, moving area would exist animated designs as illustrated in Figure 3.

To identify the area for animation, the camera’s frustum edges and camera-projected render were baked to a UV texture for each scene, placed upon the full-wall geometry and viewed in context. The subsequent floral graphics were animated in Nuke. Based on the color keys, the design elements in each shot were divided into RGB channels for 2D output. These channels were later used in Disney’s XGen to drive different properties and to control the final material colors. The animation was previewed as a UV texture in Houdini and shown to the directors through the frustum frame. Once approved, it was then baked as an animated PTex texture sequence in Houdini for consumption by Disney’s XGen to generate the animated flower instances.

For the pivotal shot of Isabela’s giant figure appearing on the floral wall behind her, as illustrated by Figure 2, projection techniques were used in Houdini using Hyperion render outputs from a modified scene camera. This render was then projected back through the original scene camera to accomplish the crash zoom



Figure 3: Shot-by-Shot animated images of the sequence

projection effect when the camera pulled back. The render outputs were processed further in a Houdini shader and run through the same texture baking process to get the various passes into the PTex sequences in order to drive different procedural effects. The final render of this jumbo floral display is shown in Figure 4.

2.2 Look Development using Disney’s XGen

Look Development was responsible for the procedural generation of the flowers, as well as creating the colors and materials. Via Disney’s XGen and the scripting language SeExpr, we were able to use the animated textures provided by Effects department to drive different properties on the procedural instances, such as color, scale, normal offset, animated noise effects, as well as the offset of the flowers from the base geometry. The workflow consisted of generating a set of animated textures on a shot-by-shot basis. These maps were formatted to be consistent across the sequence, so that the various XGen expressions and materials would automatically read in any given shot’s baked textures. Because of the unique qualities of this effect, we went through many iterations to find the right combination of properties for offset, motion, scale, and color. The procedural processes and fast preview made the iterations between the departments easier.

3 RESULTS

Final renders of the floral display are shown in Figure 1 and Figure 4. Note the huge scale of the floral walls compared to the characters as shown in the palm tree image from Figure 1 and the giant floral version of Isabela on the wall in Figure 4. In terms of scene complexity, on average, the jumbo floral display featured approximately 400,000 instances of flower elements and took 39 seconds to generate. Total of 5 unique flower element types were used amongst all of the instances.

4 CONCLUSIONS AND FUTURE WORK

We have presented techniques to bring the giant floral display to life. This close cross-department collaborations among Production Design, Effects, Look Development and Technology was vital. By utilizing existing tools and workflows in atypical and creative ways, we were able to provide a simple solution to a complex sequence efficiently and aesthetically. Future work includes to exploring ways to support animated texture coordinates in Disney’s XGen.



Figure 4: Final render of the jumbo floral display